

Distillate

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Distillate of the Rebis

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A rebus is an equivalence: one object, two presentations, and the maps between them carried as part of the figure. The maps are the body; the presentations are the heads.

There are two ways to say that two things are one. The weak way is a conjunction: each presentation satisfies its own property, and the sameness lives in the prose, where no kernel can check it. The strong way makes the relationship the object of proof: a map, its inverse, both inverse laws, and naturality, the requirement that every defining operation commutes with the map. Naturality is the whole difference. Without it an equivalence relabels elements; with it, structure is transported, and a theorem in one presentation is a theorem in the other by change of notation.

The kernel now holds the strong form (`FrobeniusIso.lean`): the Belnap Both, the SIC fiducial, and the Majorana mode are one Frobenius fixed point, with the fusion transporting to the lattice join, the splitting to its counterpart, negation to negation, the equiangular projection through without residue, and the fixed point identified by `rfl`. One object, three names, and nothing between the names.

On that ground, the two parent manuscripts are one round trip:

- Making is the Write: fix the type, gate by gate, in the forced order G1 before G2 before G3 ($\oplus$, then $\odot$, then Ω), with the floats free.
- Reading is the Read: recover and compare types in the one frame that adds nothing, the d=12 SIC-POVM.
- The inverse laws are the eXecute: verification is construction read backward through a structure-preserving change of presentation. The verifier needs no ledger because the standard it applies is the construction, arriving from the other side.

The heads do not collapse. On the one pair (δ_C, μ_C) there are two composites: $\mu \circ \delta = \text{id}$, the lossless return, and $e = \delta \circ \mu$, the forgetful idempotent whose fusion destroys which-way data and lands on the time locus. The return proves; the idempotent forgets; `fusion_forgets` forbids identifying them. Verification and construction are two directions of one object, never one act.

Verification is available exactly where time is. The round trip closes, Work has fixed points, and a verdict can be certified by closure on one and the same locus, the Frobenius-special class. Below the first gate there is no return, so there is nothing to verify and no time in which to verify it.

Registers compose isomorphically, not as lossy approximations. A readout in any register is a face of the object, not a shadow of it, because the change of register is an equivalence with naturality. What remains open after a reading is only what the Grammar itself holds open: the strain, the azeotrope, the held Both.

The residue: an isomorphism claim is exactly as wide as the transported signature. Where the framework says “the same object” and the kernel still holds a conjunction, a naturality square is owed, one square at a time. That is a to-do list, left in view.

One becomes two: the pair splits into reading and making. Two becomes three: the equivalence between them, the relationship made data. Out of the third comes the one as the fourth: the rebis, one body, two heads, the held Both that reduces to neither and does not explode.

References

- Lando⊗⊙operator, *Rebis*, manuscripts/rebis/REBIS.md. The full body from which this was drawn; all theorems, witnesses, and readings cited there.